P510/2
PHYSICS
PAPER 2
(Practical)
July/August 2026
3 hours

WAKISSHA JOINT MOCK EXAMINATIONS

Uganda Advanced Certificate of Education

PHYSICS

Paper 2

3 hours

INSTRUCTIONS TO CANDIDATES:

- *This examination paper consists of* **two** *compulsory items.*
- *Use* **blue** *or* **black** *ink. Any work done in pencil* **except** *graphs and diagrams will* **not** *be scored.*
- *Scores are awarded mainly for a clear record of the observations actually made, for their suitability, accuracy and for the use made of them.*
- *Details on the question paper should not be repeated in the responses.*
- *Graph paper is provided.*
- *Silent non-programmable scientific calculators may be used.*

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Item 1

A battery manufacturing company produces dry cells intended for use in portable electronic devices. Of recent, the company has retired its quality control scientist upon attaining the mandatory retirement age of 60 years. While the human resource department is in the process of recruiting a replacement, production has continued and quality assurance procedures must still be conducted to ensure that the dry cells produced meet the company's standards.

According to the company's quality control guidelines, a fresh dry cell should have an electromotive force (e.m.f) in the range of 1.40 V to 1.60 V and an internal resistance between 0.5 Ω and 1.2 Ω. However, in the absence of a qualified quality control scientist, the company has been unable to verify whether the dry cells currently being produced meet these requirements.

Consequently, the manager of the company has approached you with a sample dry cell from one of the production batches and other apparatus required for investigation and has requested that you carry out the investigation using two independent methods.

One method involves the use of a voltmeter and an ammeter connected in a circuit with the sample dry cell labelled X and different lengths of a bare wire labelled W, and then through analysis of the results obtained graphically you obtain the electromotive force (E) and internal resistance (r).

The second method involves the use of a slide wire potentiometer and a standard resistor of resistance 5 Ω.

Task

Carryout a scientific investigation to assess the quality of the sample dry cell provided to you by determining its electromotive force, E and internal resistance, r.

Item 2

A severe storm wasted away the pedestrian foot bridge across a river, connecting between the village and a nearby trading Centre. Concerned about the safety and mobility of residents, the L.C.1 chairperson called for an emergency community meeting to discuss possible solutions.

During the meeting, a local engineer suggested using hard wood obtained from available trees in the village to construct a temporary foot bridge.

However, he explained that the suitability of the wood depends not only on its size but also on its mechanical properties such as Young's modulus.

For the bridge to safely support people of different weights, the wood should have a Young's modulus, E, between 5.0×10^9 and 1.85×10^{10} pa. He further explained that the greatest bending stress and strain would occur when a person stands at the Centre of the bridge. Therefore if the bridge can safely withstand the load at its centre, it is likely to be safe for normal use.

The chairman L.C.1 was uncertain about how the Young's modulus of the wood could be determined. A physics student who had recently completed senior six suggested that the property could be determined experimentally using a wooden metre rule made from the same type of wood as intended for constructing the bridge.

You are provided with a wooden metre rule made from the same type of wood as that intended for constructing the bridge and the other apparatus required for the experiment.

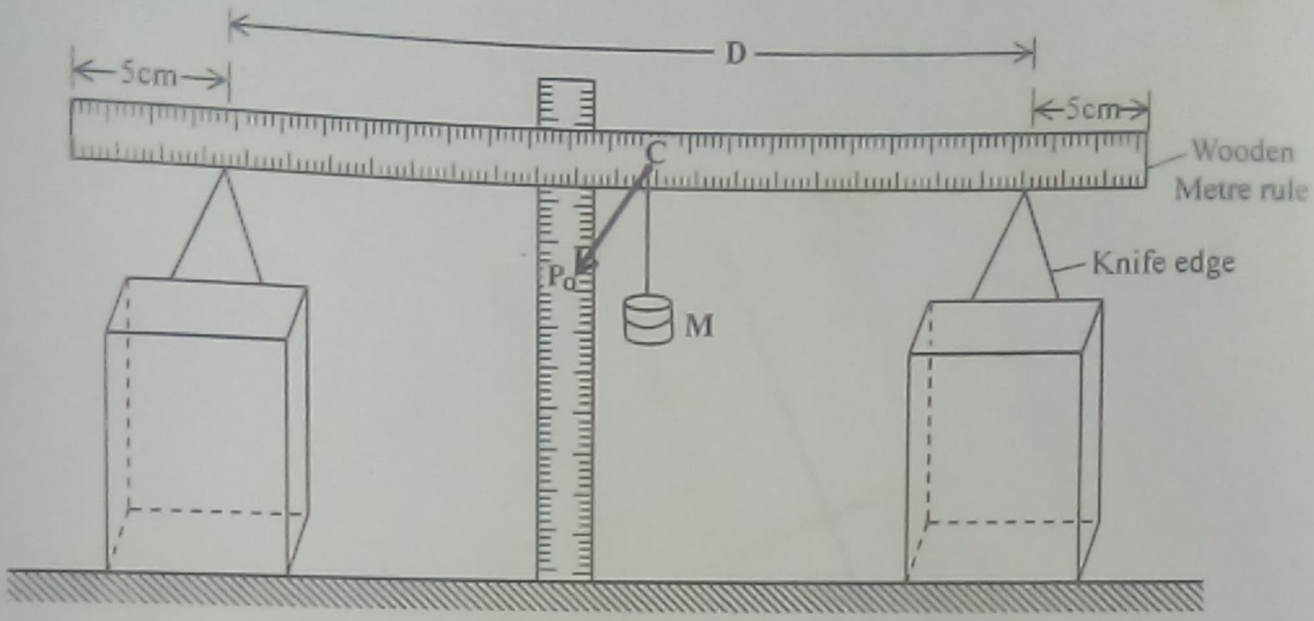
Task:

Follow the instructions provided and use the results to assess the ability of the wood to safely support people of different weights by determining its Young's modules.

(a) Measure the breadth, b, and the thickness, d, of the wooden metre rule provided.

(b) Balance the metre rule on the knife edge and note the balance point C.

(c) Arrange the apparatus as shown in the figure below.



Read and record the initial position of the pointer, P_o on the scale of the half metre rule.

(d) Attach a mass M = 0.100 kg at point C of the metre rule.

(e) Read and record the new position of the pointer P_1.

(f) Calculate the depression, y produced in the metre rule.

(g) Repeat steps (d) to (f) for values of M = 0.200, 0.300, 0.400, 0.500 and 0.600 kg.

(h) Record your results in a suitable table.

(i) Plot a graph of y against M.

(j) Determine the slope, S, of the graph.

(k) State the possible sources of errors and the corresponding mitigations to minimize / overcome them.

(l) Determine value of Young's modulus, E, from the expression $E = \frac{gD^3}{4Sbd^3}$ where $g = 9.81\ ms^{-2}$

END